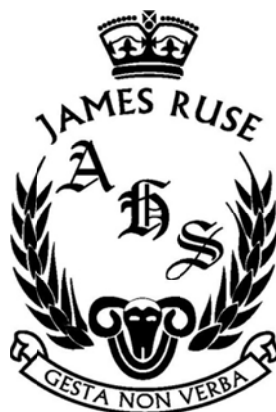


Name:	
Class:	



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2012

MATHEMATICS EXTENSION 1

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 2 hours.
- Write in black or blue pen.
- Board approved calculators & templates may be used
- A Standard Integral Sheet is provided.
- In every question, show all necessary working
- Marks may not be awarded for careless or badly arranged working.

Total Marks 70

Section I: 10 marks

- Attempt Question 1 – 10.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section.

Section II: 60 Marks

- Attempt Question 11 - 14
- Answer on blank paper unless otherwise instructed. Start a new page for each new question.
- Allow about 1 hours & 45 minutes for this section.

The answers to all questions are to be returned in separate *stapled* bundles clearly labelled Question 11, Question 12, etc. Each question must show your Candidate Number.

- 5 What are the solutions to the equation $e^{6x} - 7e^{3x} + 6 = 0$?
- (A) $x = 1$ and $x = 6$ (C) $x = 0$ and $x = \frac{\ln 6}{3}$
(B) $x = 0$ and $x = \frac{\ln 6}{2}$ (D) $x = 1$ and $x = \frac{\ln 6}{2}$
- 6 A particle moving in a straight line obeys $v^2 = -x^2 + 2x + 8$ where x is its displacement from the origin in metres and v is its velocity in ms^{-1} . The motion is simple harmonic. What is the amplitude?
- (A) 2π metres (C) 8 metres
(B) 3 metres (D) 9 metres
- 7 How many distinct permutations of the letters of the word 'ATTAINS' are possible in a straight line when the word begins and ends with the letter T?
- (A) 60 (C) 360
(B) 120 (D) 1260
- 8 If $f(x) = e^{x+2}$ what is the inverse function $f^{-1}(x)$?
- (A) $f^{-1}(x) = e^{y-2}$ (C) $f^{-1}(x) = \log_e x - 2$
(B) $f^{-1}(x) = e^{y+2}$ (D) $f^{-1}(x) = \log_e x + 2$
- 9 What is the coefficient of x^5 in the expansion of $(1 - 3x + 2x^3)(1 - 2x)^6$?
- (A) -792 (B) -720
(C) 120 (D) 312
- 10 A die is tossed 3 times. What is the probability of 0 or 1 six turning up?
- (A) $\frac{2}{27}$ (B) $\frac{25}{27}$
(C) $\frac{91}{216}$ (D) $\frac{125}{216}$

SECTION II EXTENDED RESPONSE (60 marks)**Total Marks is 60****Attempt Question 11 – 14.****Allow approximately 1 hour & 45 minutes for this section.**

Answer all questions, starting each new question on a new sheet of paper with your **student ID number** in the top right hand corner and the question number on the left hand side of your paper. All necessary working must be shown in each and every question.

QUESTION 11 (15 Marks)**Marks**

(a) Solve $\frac{4}{3x+1} < 5$. **2**

(b) If α, β and γ are the roots of the equation $x^3 + 2x^2 - 3x - 5 = 0$, **2**
find the value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$.

(c) Use the substitution $x = u^2 + 1$ for $u > 0$ to evaluate the integral: **4**

$$\int_1^5 (x+1)\sqrt{x-1} \, dx$$

(d) A series is given by $1 + \frac{1-p}{p} + \left(\frac{1-p}{p}\right)^2 + \dots$, where p is positive.

(i) Find the domain of p such that the series has a sum to infinity. **2**

(ii) Find this sum to infinity in terms of p . **1**

(e) Prove that the tangent to a parabola $x^2 = 4ay$ at a given point $P(2ap, ap^2)$ **4**
is equally inclined to the axis of the parabola and the focal chord through the point.

QUESTION 12 (15 Marks) START A NEW PAGE

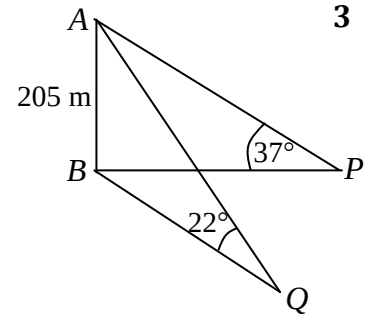
Marks

- (a) Solve: $x^3 + 2x^2 - 5x - 6 = 0$ **2**
- (b) When the polynomial $P(x)$ is divided by $x^2 - 1$, the remainder is $3x + 1$. **2**
What is the remainder when $P(x)$ is divided by $x + 1$?
- (c) In how many ways can 4 men and 4 women be arranged around a circular table if:
- (i) All women sit together? **2**
- (ii) All the men are in pairs separated by two pairs of women? **2**
- (d) Find the general solution to: $\cos 5\theta - \cos 2\theta = 0$ **2**
- (e) A thin-walled cone-shaped cup is to hold $36\pi \text{ cm}^3$ of water when full. **5**
What dimensions will minimize the amount of material needed for the cup?
[You may make use of the formula $A = \pi rs$, where s is the slant height of a cone]

QUESTION 13 (15 Marks) START A NEW PAGE

Marks

- (a) A is 205 metres above the horizontal plane BPQ .
 AB is vertical. The angle of elevation of A from P is 37°
and the angle of elevation of A from Q is 22° .
 P is due East of B and Q is South 47° East from B .


3

Calculate the distance from P to Q , to the nearest metre.

NOT TO SCALE

- (b) Use mathematical induction to show:

4

$$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$

for $n \geq 2$ where n is an integer.

- (c) A particle moves in SHM on a horizontal line and its acceleration is
 $\frac{d^2x}{dt^2} = 36 - 9x$, where x is the displacement after t seconds.

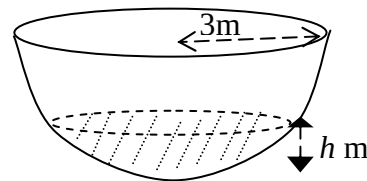
- (i) Find the centre of its motion.

1

- (ii) If the particle is initially at rest at $x = 6$, find the amplitude.

1

- (d) A hemi-spherical bowl has a radius of 3m. Oil is poured in at a constant rate of $\frac{\pi}{3} \text{ m}^3/\text{min}$.



NOT TO SCALE

- (i) Show that, when the depth of the oil is h metres, the volume of oil is:

2

$$V = \frac{\pi}{3}(9h^2 - h^3)\text{m}^3$$

- (ii) How deep is the oil after 8 minutes?

2

- (iii) At what rate is h increasing at this time?

2

QUESTION 14 (15 Marks) START A NEW PAGE

Marks

- (a) A particle is moving in a straight line and its position x , in metres, from the origin O at time t seconds is given by

$$x = 3 \cos 2t + 4 \sin 2t + 2.$$

- (i) Express $3 \cos 2t + 4 \sin 2t$ in the form **2**

$$R \cos(2t - \alpha) \quad \text{where } 0 < \alpha < \frac{\pi}{2} \text{ and } R > 0.$$

- (ii) Prove that the particle is undergoing simple harmonic motion. **2**
Find the amplitude of the motion.

- (iii) Find the maximum speed of the particle. **2**
When does the particle **first** reach this maximum speed?
Provide your answer to 2 decimal places.

- (b) Given the binomial expansion of

$$(1 + x)^n = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \text{ and}$$

$$(1 + x)^{n+1} = b_0 + b_1x + b_2x^2 + \dots + b_{n+1}x^{n+1}$$

- (i) Find the relationship for co-efficient b_k in terms of a_r . **1**

- (ii) Hence find the expression, in terms of n only, of: **3**

$$\frac{1}{a_0 a_1 \dots a_n} \times (a_0 + a_1)(a_1 + a_2) \dots (a_{n-1} + a_n) \quad \text{for } n = 1, 2, 3 \dots$$

- (c) (i) Show that $\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1}\left(\frac{2}{n^2}\right)$ for $n \geq 1$. **2**

- (ii) Hence or otherwise show that: **3**

$$\sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) = \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) + \frac{3\pi}{4}$$

END OF PAPER

Section I

10 Marks

Attempt Question 1 – 10.

Allow approximately 15 minutes for this section.

Use the multiple choice answer sheet below to record your answers to Question 1 – 10.

Select the alternative: A, B, C or D that best answers the question.

Colour in the response oval completely.

Sample:

$2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, draw a cross through the incorrect answer and colour in the new answer

ie A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word “correct” and draw an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct
 A ☒ B ☒ C ☐ D ☐

Trial HSC Examination
Mathematics Extension 1, 2012

Multiple Choice Answer Sheet

Student id number:

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Completely colour in the response oval representing the most correct answer.

1	A	☐	B	☐	C	☒	D	☐
2	A	☐	B	☐	C	☐	D	☒
3	A	☐	B	☐	C	☒	D	☐
4	A	☐	B	☐	C	☒	D	☐
5	A	☐	B	☐	C	☒	D	☐
6	A	☐	B	☒	C	☐	D	☐
7	A	☒	B	☐	C	☐	D	☐
8	A	☐	B	☐	C	☒	D	☐
9	A	☒	B	☐	C	☐	D	☐
10	A	☐	B	☒	C	☐	D	☐

Mark: /10

1/5

MATHEMATICS Extension 1 : Question 11

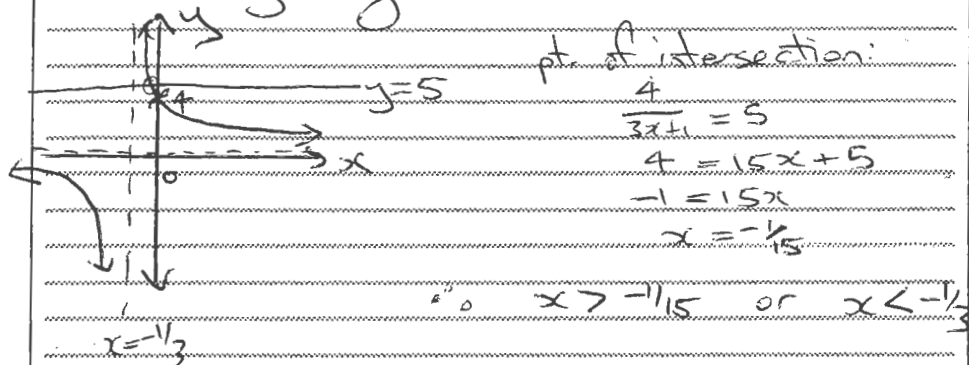
Suggested Solutions

Marks

Marker's Comments

(a) $\frac{4}{3x+1} < 5$

could have been done using a graph
or by algebra.



* 1/2 mark off if the signs were wrong

* If you only got $x > -\frac{1}{5}$, you scored one mark only.

* A lot of students were confused with the signs!

or using algebra

$(3x+1)^2 = 4$
 $\frac{4}{3x+1} < 5 \quad (3x+1)^2$

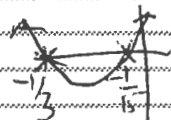
METHOD II:

$4(3x+1) < 5(9x^2 + 6x + 1)$
 $12x + 4 < 45x^2 + 30x + 5$

$-45x^2 - 18x - 1 < 0$

$45x^2 + 18x + 1 > 0$

$(15x+1)(3x+1) > 0$



$\therefore x < -\frac{1}{5} \text{ or } x > -\frac{1}{5}$

(b) $\alpha + \beta + \gamma = -\frac{b}{a} = -2$

$\alpha\beta\gamma = -\frac{d}{a} = 5$

$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma}$
 $= -\frac{2}{5}$

1/2

1/2

1/2

1/2

(c) $x = u^2 + 1$

$\frac{dx}{du} = 2u$

$dx = 2u du$

1/2

2/5

MATHEMATICS Extension 1 : Question ... 11 ...

Suggested Solutions	Marks	Marker's Comments
when $x=1, u=0$ $x=5, u=2$ $\int_1^5 (x+1)\sqrt{x-1} dx = \int_0^2 (u^2+1+1)\sqrt{u^2+1-1} \cdot 2u du$ $= \int_0^2 (u^2+2)u \cdot 2u du$ $= 2 \int_0^2 (u^4 + 2u^3) du$ $= 2 \left[\frac{u^5}{5} + \frac{2u^4}{4} \right]_0^2$ $= 2 \left(\frac{32}{5} + \frac{2 \times 8}{3} - 0 \right)$ $= \frac{64}{5} + \frac{32}{3}$ $= \frac{392}{15}$ $= 23 \frac{7}{15}$	1/2 1/2 1/2 1/2 1/2 1/2 1/2	1/2 mk 1/2 1/2 1/2 1/2 1/2 * Answers of "64/5" or "8" → 3 mks max.
(d) (i) For there to be a sum to infinity $ r < 1$ ie $-1 < r < 1$ $-1 < \frac{1-p}{p} < 1$ or $\frac{1-p}{p} < 1$ $1-p < p$ $1 < 2p$ $\frac{1}{2} < p$ $\therefore p > \frac{1}{2}$ $p > 0$ (data) $\therefore p > \frac{1}{2}$	1/2 mk 1/2 mk	* lost 1mk if they wrote $0 < r < 1$ or if they wrote $r < 1 \Rightarrow 1mk$. * lost one mark if they squared the absolute value expression or $\left \frac{1-p}{p} \right < 1$ $\frac{ 1-p }{ p } < 1$ $p \neq 0$ $1-p < p$ etc

MATHEMATICS Extension 1 : Question ... 11 ...

Suggested Solutions	Marks	Marker's Comments
d(ii) $S_{\infty} = \frac{a}{1-r}$ $= \frac{1}{1 - \frac{1-p}{p}}$ $= \frac{p}{p - (1-p)}$ $= \frac{p}{p - 1 + p}$ $= \frac{p}{2p-1}, \quad p > \frac{1}{2}$	$\frac{1}{2}$ $\frac{1}{2}$	Orks if they got the formula wrong!!
(e) <u>METHOD ONE</u> when $x = 2ap, y = ap^2$ $\frac{dy}{dx} = p$ eqn of tangent is: $y - ap^2 = p(x - 2ap)$ $y = px - ap^2$ y intercept is Q, Q is $(0, -ap^2)$ $d_{FQ} = ap^2 + a$ $= a(p^2 + 1)$ $d_{FP} = \sqrt{(2ap - 0)^2 + (ap^2 - a)^2}$ $= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2}$ $= \sqrt{a^2p^4 + 2a^2p^2 + a^2}$ $= a\sqrt{p^4 + 2p^2 + 1}$ $= a\sqrt{(p^2 + 1)^2}$ $= a(p^2 + 1)$ $\therefore d_{FP} = d_{FQ}$ (both $a(p^2 + 1)$)	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	Focus-directrix defn $FP = PM$ $= ap^2 - (-a)$ $= ap^2 + a$

MATHEMATICS Extension 1 : Question 11....

Suggested Solutions

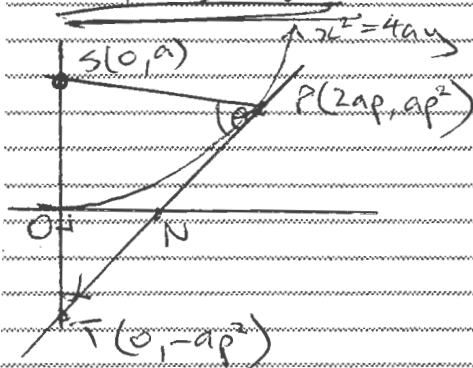
Marks

Marker's Comments

$\therefore \angle FQP = \angle FPQ$ (equal angles are opposite equal sides in $\triangle FQP$)

$\therefore$ the tangent is equally inclined to the focal chord and the axis } $\frac{1}{2}$

METHOD 2:



gradient of tangent $= p$

$$\begin{aligned} \text{gradient of focal chord} &= \frac{ap^2 - a}{2ap - 0} \\ &= \frac{a(p^2 - 1)}{2ap} \\ &= \frac{p^2 - 1}{2p} \end{aligned}$$

$\frac{1}{2}$ mk for both correct gradients

In $\triangle SPT$,

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{p - \left(\frac{p^2 - 1}{2p} \right)}{1 + p \left(\frac{p^2 - 1}{2p} \right)} \right| \end{aligned}$$

$$= \left| \frac{2p^2 - (p^2 - 1)}{2p + p(p^2 - 1)} \right|$$

$$= \left| \frac{p^2 + 1}{p^3 + 2p - p} \right|$$

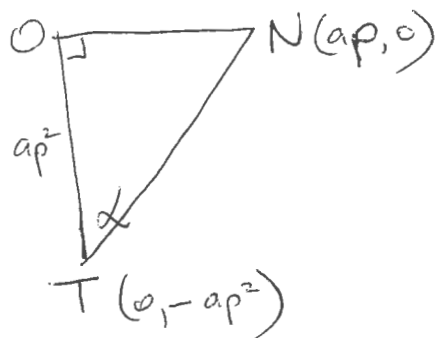
$$= \left| \frac{p^2 + 1}{p^3 + p} \right|$$

$$= \left| \frac{p^2 + 1}{p(p^2 + 1)} \right| = \frac{1}{p}$$

$\frac{1}{2}$ mk for a decent diagram with data marked on it!!

5/5

the gradient of the y-axis is undefined



In $\triangle ONT$,

$$\tan \alpha = \frac{ap}{ap^2}$$

$$\therefore \tan \alpha = \frac{1}{p}$$

1mk

$$\therefore \tan \alpha = \tan \theta = \frac{1}{p}$$

conclusion

1/2mk

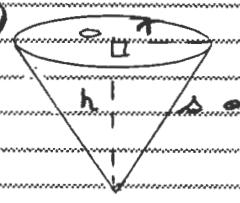
* Fudging resulted in a maximum of 2mks.

* Students were making statements without any justifications, no evidence $\therefore$ they lost marks!!!

MATHEMATICS Extension 1 : Question 12

Suggested Solutions	Marks	Marker's Comments
(a) Let $P(x) = x^3 + 2x^2 - 5x - 6$ possible integer/rational zeros $\pm 1, \pm 2, \pm 3, \pm 6$ TEST: $x-1$: $P(1) = 1+2-5-6 = -8 \neq 0$ $x+1$: $P(-1) = -1+2+5-6 = 0$ $\therefore x+1$ is a factor // $x = -1$ is a root (Factor Thm) $\begin{array}{r} x^2 + x - 6 \\ x+1 \overline{) x^3 + 2x^2 - 5x - 6} \\ \underline{x^3 + x^2} \\ 0 x^2 - 5x \\ \underline{x^2 + x} \\ 0 - 6x - 6 \\ \underline{-6x - 6} \\ 0 \end{array}$ $\therefore (x+1)(x^2 + x - 6) = (x+1)(x+3)(x-2) = 0$ $\therefore x = -3, -1 \text{ or } 2$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	$\pm 1, \pm 2, \pm 3, \pm 6$ (Factor Thm) $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ For either -3 or 2 $-\frac{1}{2}$ if +1 or +3
(b) $P(x) = A(x)Q(x) + R(x)$ $= (x^2 - 1)Q(x) + 3x + 1$ $x+1$ $PC(-1) = 0 \times Q(-1) + 3(-1) + 1$ $PC(-1) = -2$ $\therefore$ the remainder is -2 when divide by $x+1$ (Rem. Thm)	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	$\frac{1}{2}$ $\frac{1}{2}$
(c) (i) Men 4M together and 4W together Women $n(E) = 4! \times 4! = (4!)^2 = 576$ ways (ii) Method I: $(2 \times 3!) \times 4!$ $n(E) = (2 \times 3!) \times 4! = 288$ Method II $n(E) = \left(\frac{4!}{2!} \times 2! \right) \times 4! = 288$ $\uparrow$ $1^{st} 2M \text{ other}$ Method III $n(E) = 3 \times 2! \times 2! \times 4! = 288$ $\uparrow$ who pair with M Method IV $n(E) = 2 \times 2^4 \times 3 \times 3 = 288$ $\uparrow$ $\text{opposites in 2 group}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	$\frac{2!}{2!} \times 4! \times 4!$ $\frac{5!}{5} \times 4!$ $(4W, M_1, M_2, M_3, M_4)$ For '2x3!' or equiv. $\frac{1}{2} 4! \checkmark$ $\frac{1}{2} 288$ Very UNCLEAR ARGUMENTS by Students

MATHEMATICS Extension 1 : Question 12

Suggested Solutions	Marks	Marker's Comments
(d) $\therefore \cos 5\theta = \cos 2\theta$ <u>METHOD I</u> $5\theta = 2m\pi \pm 2\theta$ $\theta = \begin{cases} \frac{2m\pi}{3}, m \in \mathbb{Z} \\ \frac{2m\pi}{3}, m \in \mathbb{Z} \end{cases}$ <u>METHOD II</u> $2\theta = 2n\pi \pm 5\theta$ $\theta = \begin{cases} \frac{2n\pi}{3}, n \in \mathbb{Z} \\ -\frac{2n\pi}{3}, n \in \mathbb{Z} \end{cases}$		Expansions  2 $\frac{1}{2}$ $\frac{1}{2}$ each
(e)  $V = \frac{1}{3}\pi r^2 h = 36\pi$ $r^2 h = 108$ $h = \frac{108}{r^2}$ $s^2 = r^2 + h^2$ (Pyth. Thm.) $s^2 = r^2 + \frac{108^2}{r^4} = \frac{r^6 + 108^2}{r^4}$ $s = \frac{\sqrt{r^6 + 108^2}}{r^2}$ Now $A(r) = \pi r s$ $= \pi r \frac{\sqrt{r^6 + 108^2}}{r^2}$ $A(r) = \frac{\pi}{r} \sqrt{r^6 + 108^2}$ $\frac{dA}{dr} = \pi \left[\frac{-1}{r^2} \sqrt{r^6 + 108^2} + \frac{1}{r} \frac{6r^5}{2\sqrt{r^6 + 108^2}} \right]$ $= \pi \left[\frac{3r^4}{\sqrt{r^6 + 108^2}} - \frac{\sqrt{r^6 + 108^2}}{r^2} \right]$ For possible max/min value of A to occur $\frac{dA}{dr} = 0$ $\therefore \frac{3r^4}{\sqrt{r^6 + 108^2}} = \frac{\sqrt{r^6 + 108^2}}{r^2}$ $3r^6 = r^6 + 108^2$ $2r^6 = 108^2 = 11664$ $r^6 = 5832 = 54 \times 108 = 2 \times 3^6$ $r = \sqrt[6]{5832} = 3\sqrt{2} = 4.24 \dots$ as $r > 0$	$\frac{1}{2}$ $\frac{1}{2}$ $(\frac{1}{3})$ 1 1 $\frac{1}{2}$	$108^2 = 11664$ $s^2 = \frac{108}{h} + h^2 = \frac{108 + h^3}{h}$ $s = \sqrt{\frac{h^3 + 108}{h}} = \frac{\sqrt{h^3 + 108}}{\sqrt{h}}$ $A = \pi \sqrt{\frac{108}{h}} \sqrt{\frac{h^3 + 108}{h}}$ $A(h) = \pi \sqrt{\frac{108h + 108^2}{h^2}}$ $A'(h) = \frac{\pi}{2\sqrt{\dots}} \left(108 - \frac{2 \times 108^2}{h^3} \right)$ $= \frac{\pi}{\sqrt{\dots}} \left(54 - \frac{108^2}{h^3} \right)$ $54 = \frac{108^2}{h^3}$ $h^3 = \frac{108^2}{54} = 216$ $h = 6, (h > 0)$

MATHEMATICS Extension 1 : Question.....12

$$r = 4.24 \quad A' = -0.06$$

Suggested Solutions

Marks

Marker's Comments

CONT:

$$(e) \quad r = 3\sqrt{2} = \sqrt{18} = 4.24...$$

$$\therefore h = \frac{108}{r^2} = \frac{108}{18} = 6; [A = 18\pi\sqrt{3}]$$

TEST NATURE of $r = 3\sqrt{2} = 4.24...$

r	4	4.1	4.2	4.24...	4.25	4.3	4.5	5
$\frac{dA}{dr}$	-5.43	-3.16	-0.93	0	0.16	1.24	5.43	14.90
$\frac{d^2A}{dr^2}$	$\ominus$	$\ominus$	$\ominus$	$\ominus$	$\oplus$	$\oplus$	$\oplus$	$\oplus$
	$\backslash$	$\backslash$	$\backslash$	$-$	$/$	$/$	$/$	$/$

$\frac{1}{2}$ each

$\therefore$ a Relative min T.P at $r = 3\sqrt{2}$

And since there exist only 1 T.P for $r > 0$

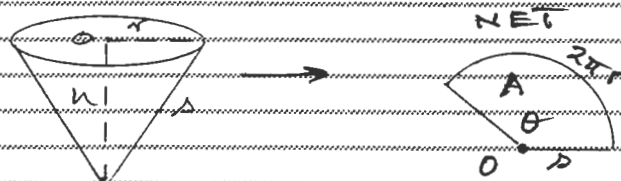
$\therefore$ absolute min Area occurs when

dimensions $3\sqrt{2}$ cm for radius and 6 cm for height

$$r = 3\sqrt{2} \quad \frac{1}{2}$$

$$h = 6$$

NOTE: Background Y10



$$\frac{\theta}{2\pi} = \frac{2\pi r}{2\pi s} = \frac{A}{\pi s^2}$$

$$\therefore \theta = \frac{2\pi r}{s}$$

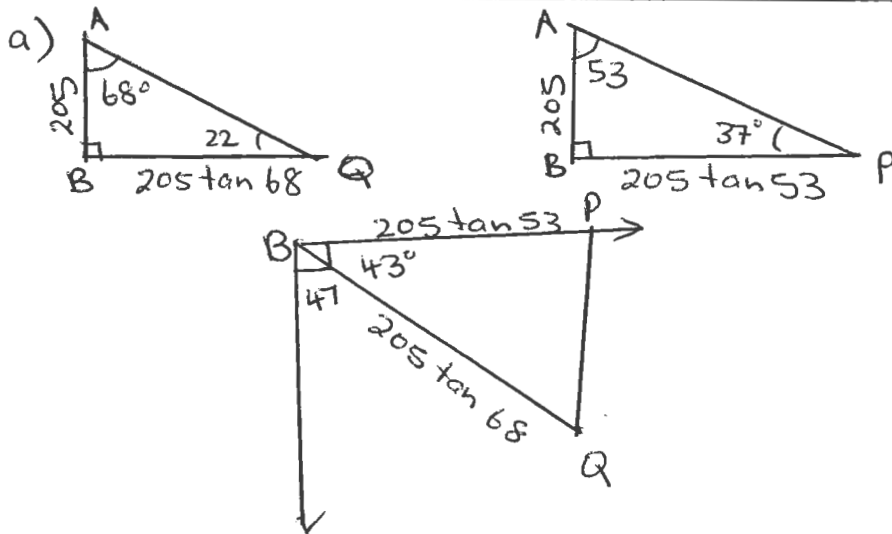
$$\text{and } A = \frac{1}{2}s^2\theta = \frac{1}{2}s^2 \times \frac{2\pi r}{s}$$

$$A = \pi r s \quad [\text{Y10 recall formula}]$$

Suggested Solutions

Marks

Marker's Comments



$$PQ^2 = (205 \tan 53)^2 + (205 \tan 68)^2 - 2(205^2 \tan 53 \tan 68) \cos 43^\circ$$

$$= 205^2 (\tan^2 53 + \tan^2 68 - 2 \tan 53 \tan 68 \cos 43^\circ)$$

$$PQ = \sqrt{205^2 (\tan^2 53 + \tan^2 68 - 2 \tan 53 \tan 68 \cos 43^\circ)}$$

$$= 359.9350248 \dots$$

$$= 360 \text{ m (nearest metre).}$$

Alternatively

BP could also be.

$$BQ = \frac{205}{\tan 22}$$

$$BP = \frac{205}{\tan 37}$$

✓ 1 mark for getting lengths of BP and BQ

✓ 1 mark calculating angle PBQ as 43°

✓ 1 mark for correct formula and working

b)

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$

Step 1 Prove true for $n=2$: $P(2)$

LHS

$$1 - \frac{1}{2^2} = 1 - \frac{1}{4}$$

$$= \frac{3}{4}$$

RHS

$$\frac{2+1}{2 \times 2}$$

$$= \frac{3}{4} = \text{LHS}$$

$\therefore$ Statement is true for $n=2$.

✓ 1 mark step 1
1 mark for 2 steps ~~2+3~~

2 step 3
Prove LHS = RHS

$\frac{1}{2}$ mark for conclusion statement must mention true for all integers $n \geq 2$
or
PMI.

Suggested Solutions

Marks

Marker's Comments

Step 2 : Assume true for $n=k$.

$$\text{i.e. } \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{k^2}\right) = \frac{k+1}{2k}.$$

$\frac{1}{2}$.

Step 3 prove true for $n=k+1$

$$\text{i.e. } \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{k^2}\right) \left(1 - \frac{1}{(k+1)^2}\right) = \frac{k+2}{2k+1}$$

Some students
have this
step missing

LHS $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{k^2}\right) \left(1 - \frac{1}{(k+1)^2}\right)$

$$= \left(\frac{k+1}{2k}\right) \left(1 - \frac{1}{(k+1)^2}\right) \text{ (by assumption)}$$

$-\frac{1}{2}$ Mark if
assumption is
not mentioned

$$= \left(\frac{k+1}{2k}\right) \left(\frac{(k+1)^2 - 1}{(k+1)^2}\right)$$

$$= \frac{k^2 + 2k + 1 - 1}{2k(k+1)}$$

$$= \frac{k^2 + 2k}{2k(k+1)}$$

$$= \frac{k^2 + 2k}{2k(k+1)}$$

$$= \frac{k+2}{2(k+1)} = \text{RHS}$$

Step 4 statement is true for
 $n=k+1$ if assumed true for
 $n=k$. Since the statement
has been proven true for $n=2$,
it is true for all integers $n \geq 2$.

or $\therefore$ by the PMI $P(n)$ is true for $n=2, 3, 4, \dots$

Suggested Solutions

Marks

Marker's Comments

c) (i) $\frac{d^2x}{dt^2} = 36 - 9x$

compare $\ddot{x} = -n^2(x - x_0)$
 $\ddot{x} = -9(x - 4)$
 $\uparrow$ centre

$\therefore$ centre is $x = 4$

ii) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 36 - 9x$
 $\frac{1}{2} v^2 = \int (36 - 9x) \cdot dx$
 $\frac{v^2}{2} = 36x - \frac{9x^2}{2} + C$

when $x = 6, v = 0$

$0 = 216 - 162 + C$

$\therefore C = -54$

Equation is $\frac{v^2}{2} = 36x - \frac{9x^2}{2} - 54$

$v^2 = 72x - 9x^2 - 108$
 $= 9(8x - x^2 - 12)$
 $= -9(x^2 - 8x + 12)$

$v^2 = -9(x - 4)^2 - 4$
 $= -9(4 - (x - 4)^2)$

compare $v^2 = n^2(a^2 - (x - x_0)^2)$

$a^2 = 4$

$a = 2 \quad (a > 0)$

OR:

write it in the form

$x = b + a \cos nt + c$
 $\uparrow$
 centre

so $x = 4 + a \cos 3t$ (from part (i))
 $n^2 = 9$
 $3\omega = 3$

when $t = 0, x = 6$

$6 = 4 + a \cos 3t$

$a \cos 3t = 2$

$a \cdot (1) = 2$

$a = 2$

✓ 1 mark

For 1 mark be careful.

If students got $6 - 4 = 2$ only they must justify and mention particle is at end point

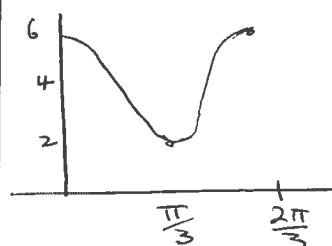
$v^2 = n^2[a^2 - (x - x_0)^2]$

$v^2 = 9[a^2 - (x - 4)^2]$

$x = 6, v = 0$

Leads to $a = 2$

Because it is initially at rest and at end of motion $c = 0$



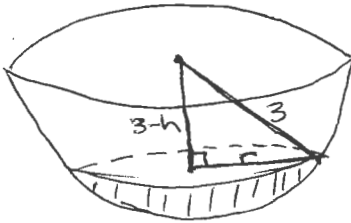
1 mark

Suggested Solutions

Marks Awarded

Marker's Comments

d) i)



Using Pythagoras

$$\begin{aligned} 3^2 &= r^2 + (3-h)^2 \\ r^2 &= 9 - (3-h)^2 \\ &= 9 - (9 - 6h + h^2) \\ &= 6h - h^2 \end{aligned}$$

$$\begin{aligned} \text{Volume} &= \pi \int_0^h (6h - h^2) dh \\ &= \pi \left[3h^2 - \frac{h^3}{3} \right]_0^h \\ &= \pi \left(3h^2 - \frac{h^3}{3} \right) \\ &= \frac{\pi}{3} (9h^2 - h^3) \text{ as required} \end{aligned}$$

Alternatively

Rotated $x^2 + y^2 = 9$ around x axis between $x = 3$ and $x = 3-h$.

$$\therefore y^2 = 9 - x^2$$

$$\begin{aligned} \text{Volume} &= \pi \int_{3-h}^3 (y^2) dx \\ &= \pi \int_{3-h}^3 (9 - x^2) dx \\ &= \left[9x - \frac{x^3}{3} \right]_{3-h}^3 \\ &= \pi \left[(27 - 9) - \left(9(3-h) - \frac{1}{3}(3-h)^3 \right) \right] \\ &= \pi \left(3h^2 - \frac{h^3}{3} \right) \\ &= \frac{\pi}{3} (9h^2 - h^3) \text{ m}^3 \end{aligned}$$

2 marks

$$\begin{aligned} x^2 &= 9 - y^2 \\ \therefore V &= \pi \int_0^h (9 - y^2) dy \end{aligned}$$

is incorrect.
limits should be 3 and 3-h.

If recognised circle is rotated with correct limits

$$\pi \int_{3-h}^3 dx$$

is 1 mark.

and 1 mark for correct subsequent working without making the question simpler.

$$\begin{aligned} &\pi \left[18 - 9 \left(\frac{3-h}{3} \right) (27 - (3-h)^2) \right] \\ &= \pi \left[\dots \right] \end{aligned}$$

Suggested Solutions

Marks Awarded

Marker's Comments

d(ii) After 8 minutes, $V = \frac{8\pi}{3}$.

$$\therefore \frac{8\pi}{3} = \frac{\pi}{3} (9h^2 - h^3)$$

$$8 = 9h^2 - h^3$$

$$h^3 - 9h^2 + 8 = 0$$

$$(h-1)(h^2 - 8h - 8) = 0$$

$$h-1=0 \quad \text{or} \quad h^2 - 8h - 8 = 0$$

$$\therefore h = 1$$

$$\frac{8 \pm \sqrt{64 - (-32)}}{2}$$

$$= \frac{8 \pm \sqrt{96}}{2}$$

$$= 4 \pm 2\sqrt{6}$$

$$\text{as } 0 \leq h \leq 3$$

$h=1$ is only solution

→ 1 mark.

→ $\frac{1}{2}$ mark

→ $\frac{1}{2}$ mark

$$\begin{array}{r} h^2 - 8h - 8 \\ h-1 \overline{) h^3 - 9h^2 + 0h + 8} \\ \underline{h^3 - h^2} \\ -8h^2 + 6h \\ \underline{-8h^2 + 8h} \\ -8h + 8 \\ \underline{-8h + 8} \\ 0 \end{array}$$

Some ended up with $h^2 - 8$ instead of $h^2 - 8h - 8$

Some tested $h=1$ on $h^3 - 9h^2 + 8$ only Did not mention other solutions, not working

d(iii) $\frac{dV}{dt} = \frac{\pi}{3}$ (given)

$$\frac{dV}{dh} = \frac{\pi}{3} (18h - 3h^2)$$

$$= \frac{\pi}{3} (6h - h^2)$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$= \frac{1}{\pi(6h - h^2)} \times \frac{\pi}{3} \quad \checkmark \text{ 1 mark.}$$

$$\frac{dh}{dt} = \frac{1}{3(6h - h^2)}$$

when $h=1$ $\frac{dh}{dt} = \frac{1}{3(6-1)} = \frac{1}{15} \quad \checkmark \text{ 1 mark.}$

$\therefore$ when oil is 1m deep, h is increasing at a rate of $\frac{1}{15}$ metres/min.

$$\begin{aligned} \frac{dV}{dt} &= \frac{dV}{dh} \cdot \frac{dh}{dt} \\ \frac{\pi}{3} &= \frac{\pi}{3} (6h - h^2) \frac{dh}{dt} \\ \Rightarrow \frac{dh}{dt} &= \frac{1}{3(6h - h^2)} \end{aligned}$$

Some forgot to invert

$$\frac{dV}{dh} = \pi(6h - h^2)$$

$$\therefore \frac{dh}{dV} = \frac{1}{\pi(6h - h^2)}$$

Some calculated the first few steps in part (ii).

Suggested Solutions	Marks	Marker's Comments
a) $x = 3 \cos 2t + 4 \sin 2t + 2$ (i) $3 \cos 2t + 4 \sin 2t = R \cos(2t - \alpha)$ $R > 0$ $0 < \alpha < \frac{\pi}{2}$ $3 \cos 2t + 4 \sin 2t = R \cos 2t \cos \alpha + R \sin 2t \sin \alpha$ $\therefore R \cos \alpha = 3$ $R \sin \alpha = 4$ $\therefore R^2 = 3^2 + 4^2$ $R = 5$ $R > 0$ $\cos \alpha = 3/5$ $\sin \alpha = 4/5$ $0 < \alpha < \frac{\pi}{2}$ $\therefore \alpha = \tan^{-1} 4/3$ or $\cos^{-1} 3/5$ or $\sin^{-1} 4/5$ $\therefore 3 \cos 2t + 4 \sin 2t = 5 \cos(2t - \tan^{-1} 4/3)$ (ii) $x = 5 \cos(2t - \tan^{-1} 4/3) + 2$ $\dot{x} = -10 \sin(2t - \tan^{-1} 4/3)$ $\frac{d\dot{x}}{dt} = -20 \cos(2t - \tan^{-1} 4/3)$ $\ddot{x} = -20 \sin(2t - \tan^{-1} 4/3)$ $\ddot{x} = -\frac{20}{5} (x - 2)$ $\ddot{x} = -4(x - 2)$ $\ddot{x} = -(2^2)(x - 2)$ $\therefore x$ is in the form $x = -n^2(x - b)$ where $n = 2$ $b = 2$ motion is SHM. Amplitude is 5 metres (iii) $x = -10 \sin(2t - \tan^{-1} 4/3)$ $\therefore$ max speed is 10 m/s. $\therefore \sin(2t - \tan^{-1} 4/3) = 1$ $2t - \tan^{-1} 4/3 = (2n+1)\frac{\pi}{2}$ $n \in \mathbb{Z}$ $\therefore$ first time $2t = \frac{\pi}{2} + \tan^{-1} 4/3$ $t = \frac{1}{2} [\frac{\pi}{2} + \tan^{-1} 4/3]$ $= 1.0249045712$ $= 1.025$ (2dp) b) (i) $b_k = a_{k-1} + a_k$ from Pascal's triangle (alternatively $b_k = \binom{n+1}{n+1-k} a_k$) (ii) $\frac{1}{a_0 a_1 \dots a_n} \times (a_0 + a_1)(a_1 + a_2) \dots (a_{n-1} + a_n)$ $= \frac{1}{a_0} \times \frac{1}{a_1 a_2 \dots a_n} \times (b_1 b_2 \dots b_n)$ $= 1 \times \frac{b_1 b_2 \dots b_n}{a_1 a_2 \dots a_n}$ $a_0 = 1$ now $\frac{b_k}{a_k} = \frac{c_k}{c_k} = \frac{(n+1)!}{(n-k+1)! k!} \times \frac{(n-k)! k!}{n!}$ $= \frac{n+1}{n-k+1}$	(2) (2) (2) (2) (2) (1) (3)	(1/2) for $R = 5$ (1/2) Both $\sin + \cos$ equations (1/2) α value. (1/2) answer. last answer only accepted. (1/2) $\ddot{x}$ (1/2) $\ddot{x}$ (1/2) reason. (1/2) amplitude Several methods. eg $\ddot{x} = 10$ $\ddot{x} = 0$ trig equation 1 solution 4 1dp. (1/2) max speed. (1) correct relationship (1/2) product of b's (1/2) $a_0 = 1$ 1 coefficients expansions 1 answer with working

Suggested Solutions

Marks

Marker's Comments

(ii) continued $b_1, b_2, \dots, b_n = \frac{(n+1)(n+1) \dots (n+1)}{(n)(n-1) \dots (1)}$
 $a_1, a_2, \dots, a_n = \frac{(n+1)^n}{n!}$

c) (i) To show $\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1}\left(\frac{2}{n^2}\right) \quad n \geq 1$
 consider LHS

Let $\tan^{-1}(n+1) = \alpha \quad \therefore \tan \alpha = n+1 \quad \frac{\pi}{4} < \alpha < \frac{\pi}{2}$

$\tan^{-1}(n-1) = \beta \quad \tan \beta = n-1 \quad 0 < \beta < \frac{\pi}{2}$

$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$
 $= \frac{(n+1) - (n-1)}{1 + (n+1)(n-1)}$
 $= \frac{2}{1 + n^2 - 1} = \frac{2}{n^2}$

$\therefore \alpha - \beta = k\pi + \tan^{-1}\left(\frac{2}{n^2}\right) \quad k \in \mathbb{Z}$

But $-\frac{\pi}{4} < \alpha - \beta < \frac{\pi}{2} \quad \therefore k = 0$

$\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1}\left(\frac{2}{n^2}\right)$

(ii) To show $\sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) = \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) + \frac{3\pi}{4}$

LHS = $\sum_{r=1}^n [\tan^{-1}(r+1) - \tan^{-1}(r-1)]$

$= \tan^{-1}2 - \tan^{-1}0 + \tan^{-1}3 - \tan^{-1}1 + \dots + \tan^{-1}n + \tan^{-1}(n-2) + \tan^{-1}(n+1) + \tan^{-1}(n-1)$
 $= \tan^{-1}(n) + \tan^{-1}(n+1) - \tan^{-1}1$

consider

$\tan(\tan^{-1}n + \tan^{-1}(n+1))$
 $= \frac{\tan(\tan^{-1}n) + \tan(\tan^{-1}(n+1))}{1 - \tan(\tan^{-1}n) \times \tan(\tan^{-1}(n+1))}$
 $= \frac{n + n+1}{1 - n(n+1)}$
 $= \frac{2n+1}{1 - n - n^2}$

$\tan^{-1}n + \tan^{-1}(n+1) = k\pi + \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) \quad k \in \mathbb{Z}$

But $\frac{\pi}{4} \leq \tan^{-1}n < \frac{\pi}{2} \quad n \geq 1$

and $\frac{\pi}{4} < \tan^{-1}(n+1) < \frac{\pi}{2}$

$\frac{\pi}{2} < \tan^{-1}n + \tan^{-1}(n+1) < \pi$
 $k = 1$

$\therefore \tan^{-1}n + \tan^{-1}(n+1) = \pi + \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right)$

$\therefore \sum_{r=1}^n \tan^{-1}\left(\frac{2}{r^2}\right) = \pi + \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) - \tan^{-1}(1)$
 $= \pi + \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) - \frac{\pi}{4}$
 $= \tan^{-1}\left(\frac{2n+1}{1-n-n^2}\right) + \frac{3\pi}{4}$

(1/2) tan expansion

(1/2) simplifying

(1/2) solution

(1/2) restrictions

(1/2) substitution

(1/2) expansion + cancelling

(1/2) tan expression and simplifying

(1/2) general solution

(1/2) correct restriction

(1/2) answer with working